IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-43

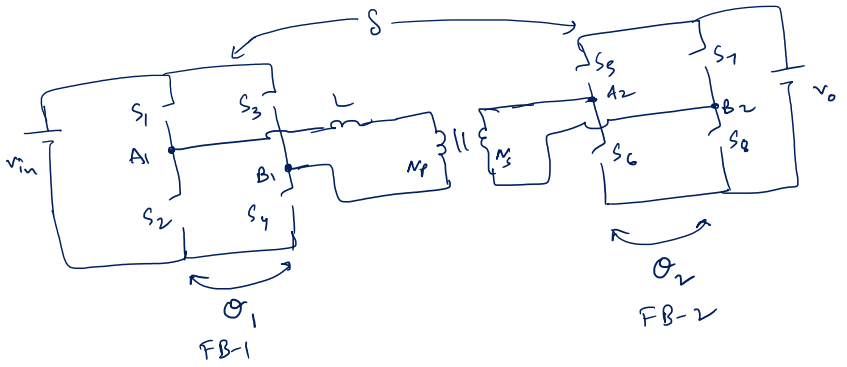
Dual Active Bridge Converter-II

Dr. Apurv Kumar Yadav

Department of Electrical Engineering



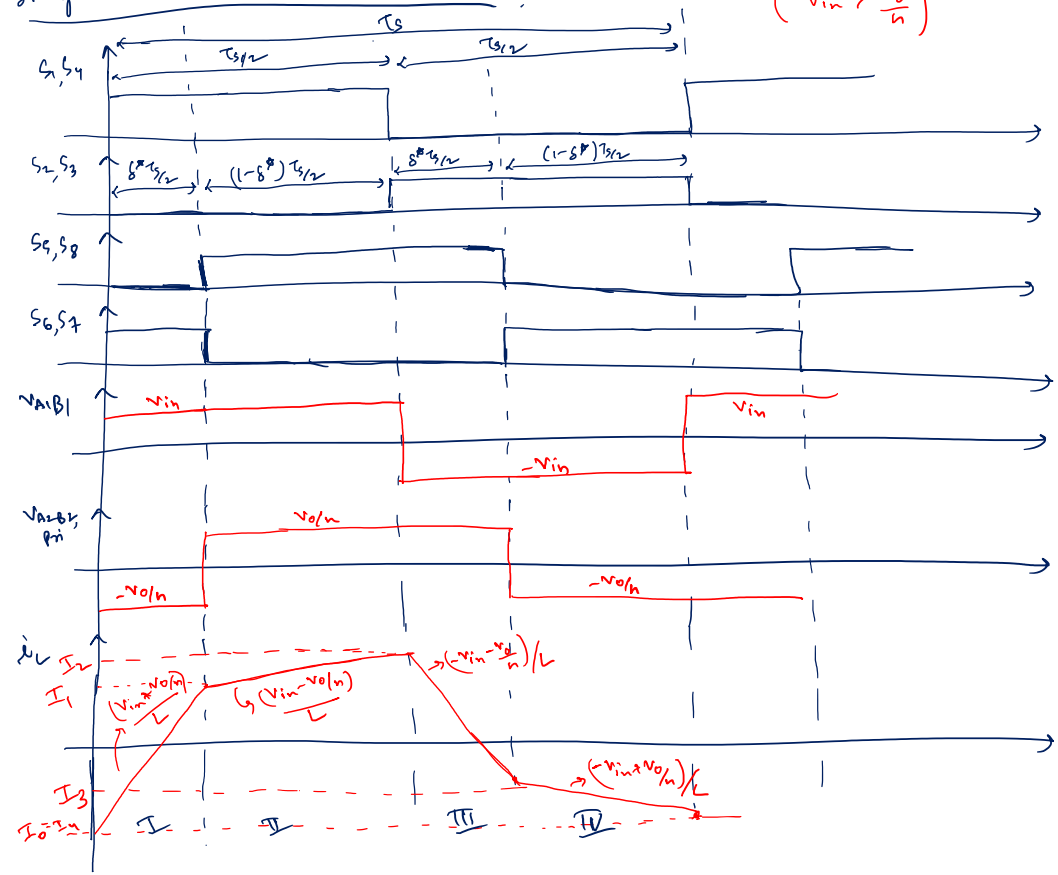
Recap



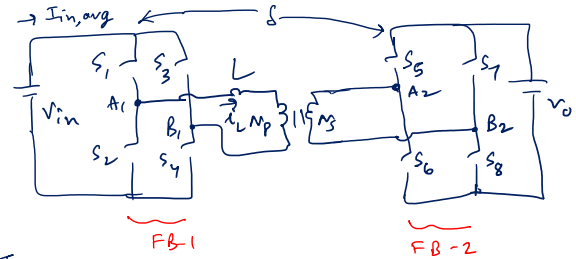
Different types of Modulation

1. Single Phase Shift (SPS) → "δ"
2. Dual Phase Shift (DPS) → "δ", θ_1 or "δ", θ_2 or θ_1, θ_2
3. Triple Phase Shift (TPS) → $\delta, \theta_1, \theta_2$
4. Triple Phase Shift with variable frequency (TPSVF) → $\delta, \theta_1, \theta_2, f_{sw}$
5. Dual Phase Shift with variable frequency → very rarely used → f_{sw}, δ, θ_2 or f_{sw}, δ, θ_1 or $\theta_1, \theta_2, f_{sw}$

Single phase shift modulation



($V_{in} > \frac{V_o}{n}$)



$360^\circ \rightarrow T_s$

$180^\circ \rightarrow T_{s/2}$

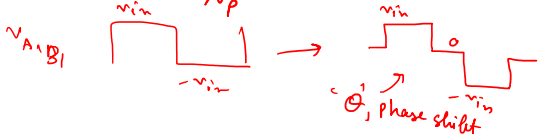
$\delta \rightarrow \frac{\delta}{180} T_{s/2}$

$\delta^* = \frac{\delta}{180}$

$\delta^* T_{s/2}$

$V_{A_2B_2,pri} \rightarrow V_{A_2B_2}$ referred to primary side

$n = \frac{N_s}{N_p}$



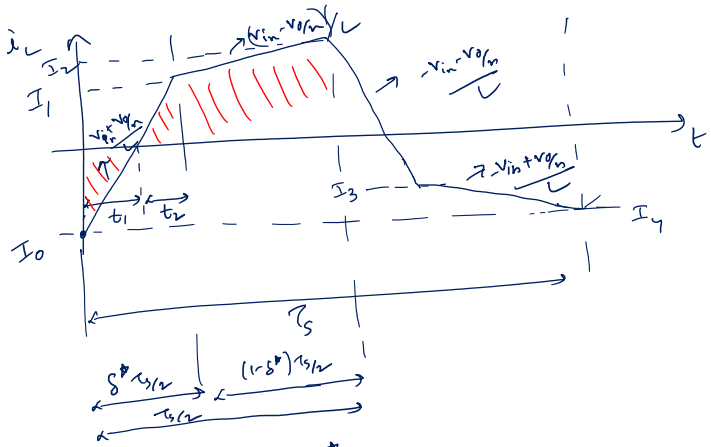
$\Rightarrow I_2 = -I_0$

$I_1 = -I_3$

$I_4 = I_0$

$\Downarrow$

This ensure no DC offset in the transformer primary current



$I_0 = -I_2 = I_4$

$I_3 = -I_1$

$\delta^* = \delta/180^\circ$ → δ → Phase shift in degrees

$t_1 + t_2 = \delta^* T_s/2$

$$\Rightarrow \frac{-I_0}{t_1} = \frac{I_1}{t_2}$$

$$\Rightarrow \frac{I_2}{t_1} = \frac{I_1}{t_2} \rightarrow (1)$$

$$\frac{L I_1}{t_2} = V_{in} + V_o \rightarrow (2)$$

$$I_1 - I_0 = \left(\frac{V_{in} + V_0/n}{L}\right) \delta^* T_s/2$$

$$I_2 - I_1 = \left(\frac{V_{in} - V_0/n}{L}\right) \frac{(1-\delta^*) T_s}{2} \longrightarrow (3)$$

$$\Rightarrow I_1 + I_2 = \left(\frac{V_{in} + V_0/n}{L}\right) \delta^* T_s/2 \longrightarrow (4) \qquad (\text{as } I_0 = -I_2)$$

$$\Rightarrow I_2 = \frac{T_s}{4L}\left[\frac{2V_0\delta^*}{n} + V_{in} - V_0/n\right] \longrightarrow (5) \qquad (\text{By solving } (3) \& (4))$$

$$I_1 = T_s/4L\left[2V_{in}\delta^* - V_{in} + V_0/n\right] \longrightarrow (6)$$

$$\Rightarrow t_1 = \frac{\delta^* T_s}{2} - t_2$$

$$\Rightarrow t_1 = \frac{\delta^* T_s}{2} - \frac{LI_1}{V_{in} + V_0/n} \qquad (\text{from eq } (2))$$

(Substitute I_1 from (6))

$$\Rightarrow t_1 = \frac{T_s}{4(V_{in} + V_0/n)}\left[2\frac{V_0}{n}\delta^* + V_{in} - V_0/n\right] \longrightarrow (7)$$

$$\Rightarrow t_2 = \frac{T_s}{4\left(V_{in}+\frac{V_o}{n}\right)}\left[2V_{in}\delta^* - V_{in} + \frac{V_o}{n}\right] \longrightarrow (8)$$

$$I_{in,avg} = \langle i_L \rangle_{T_s/2} = \frac{2}{T_s}\left[-\tfrac{1}{2} I_2 t_1 + \tfrac{1}{2} I_1 t_2 + \tfrac{1}{2}(I_1+I_2)(1-\delta^*)\frac{T_s}{2}\right]$$

from (7) $\rightarrow t_1$, (8) $\rightarrow t_2$, (5) $\rightarrow I_2$, (6) $\rightarrow I_1$

$$\Rightarrow I_{in,avg} = \frac{(1-\delta^*)\delta^* V_o}{2nL f_{sw}} = \frac{(1-\delta/180)\,\delta/180\, V_o}{2nL f_{sw}} \quad (\delta \rightarrow \text{in degrees}) \qquad \left(f_{sw} = \tfrac{1}{T_s}\right)$$

$$\Rightarrow P_{in,avg} = V_{in} \cdot I_{in,avg}$$

$$\Rightarrow P_{in,avg} = \frac{(1-\delta^*)\delta^* V_o V_{in}}{2nL f_{sw}}$$

from the power balance

$$P_{o,avg} = P_{in,avg} = P_{av}$$

$$\Rightarrow P_{av} = \frac{(1-\delta^*)\delta^* V_o V_{in}}{2nL f_{sw}} \longrightarrow (9)$$

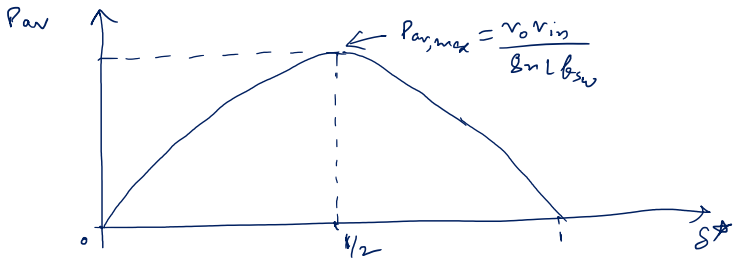
$$\Rightarrow \frac{d P_{av}}{d\delta^*} = 0$$

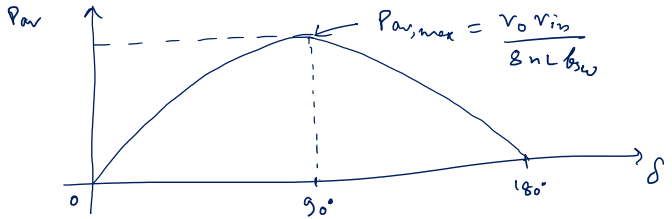
$$\Rightarrow 1 - 2\delta^* = 0$$

$$\Rightarrow \delta^* = 1/2 \quad (\Rightarrow \delta = 90^\circ)$$

$$\Rightarrow P_{av,max} = \frac{v_o v_{in} \cdot 1/2 \cdot 1/2}{2\pi L f_{sw}}$$

$$\Rightarrow \boxed{P_{av,max} = \frac{v_o v_{in}}{8\pi L f_{sw}}}$$





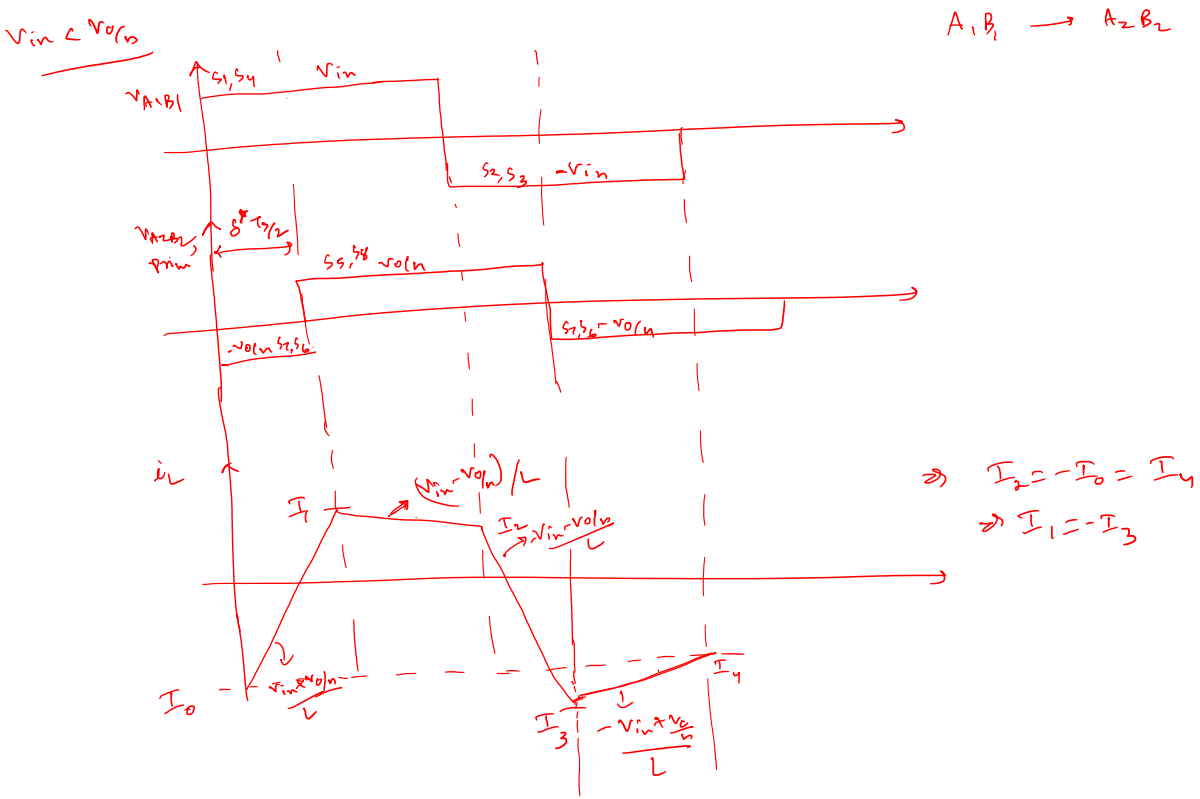
$\Rightarrow$ for the $v_{in(min)}$, $v_{o(min)}$, we can able to transfer $P_{av,max}$

$$\Rightarrow L \leq \frac{v_{in(min)}\, v_{o(min)}}{8\pi f_{sw} P_{av,max}}$$

($v_{in(min)}$, $v_{o(min)}$ → specification; f_{sw} → designer's choice; $P_{av,max}$ → specification; L → designer's choice)

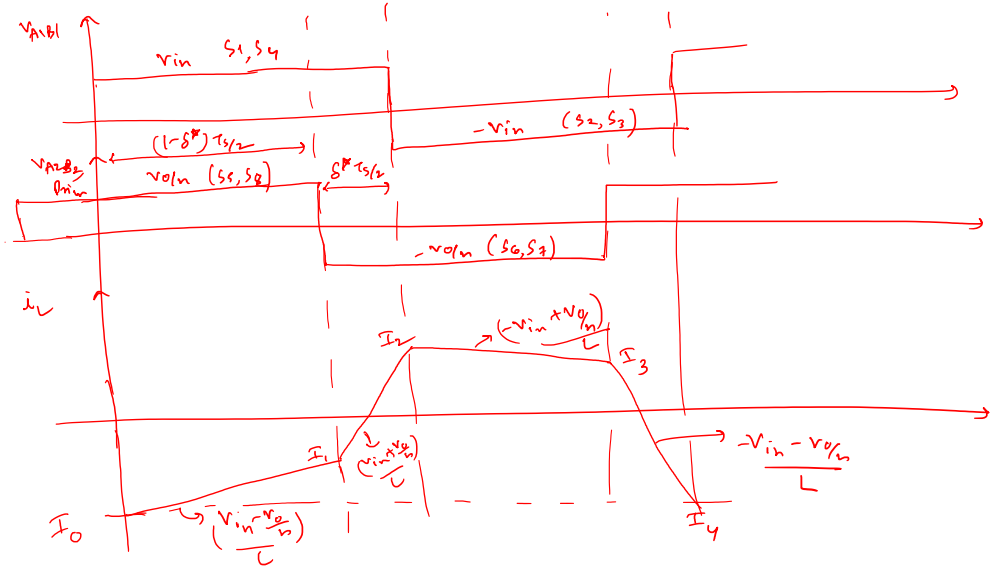
(Max value of 'L')

δ^* → "0.05 → 0.95"


$V_{in} < V_{o/n}$
$A_1B_1 \rightarrow A_2B_2$
$V_{A_1B_1}$
S_1, S_4
V_{in}
S_2, S_3
$-V_{in}$
$V_{A_2B_2}$ primary
$\delta^* T_s/2$
S_5, S_8
$V_{o/n}$
S_7, S_6
$-V_{o/n}$
$-V_{o/n}$ S_7, S_6
i_L
I_1
$(V_{in} - V_{o/n})/L$
I_2
$\frac{V_{in} - V_{o/n}}{L}$
I_0
$\frac{V_{in} - V_{o/n}}{L}$
I_4
I_3
$\frac{-V_{in} + V_{o/n}}{L}$
$\Rightarrow I_2 = -I_0 = I_4$
$\Rightarrow I_1 = -I_3$

Backward Power Flow $(v_{in} \to v_{o/n})$

$A_2 B_2 \to A_1 B_1$

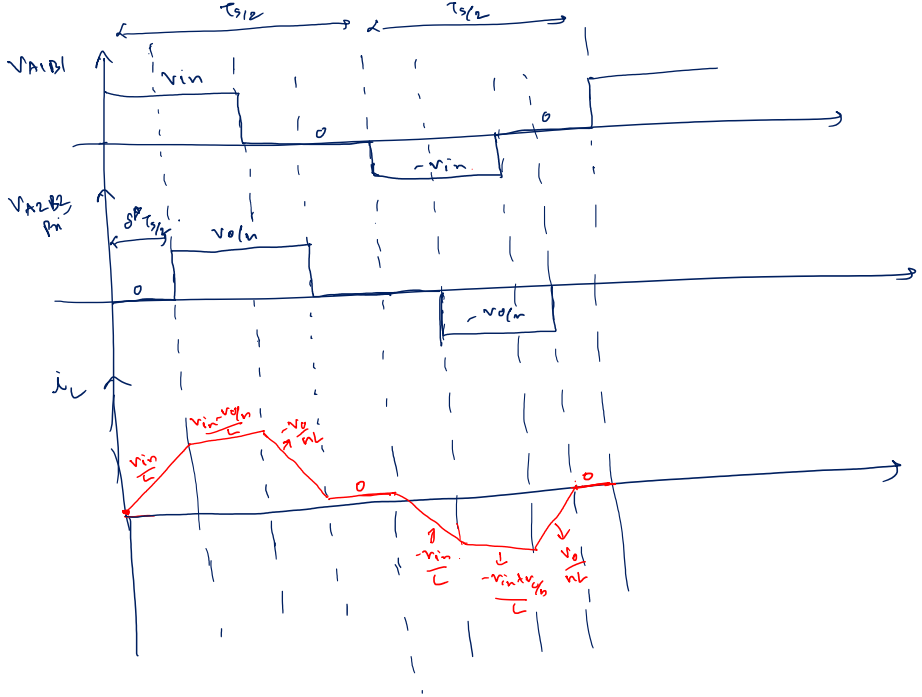


$\Rightarrow I_0 = I_4 = -I_2$

$I_1 = -I_3$

$$P_{in,avg} = \frac{-v_0 v_{in} \delta^* (1-\delta^*)}{2 n L f_{sw}}$$

TPS modulation $\left(v_{in} > \frac{v_o}{n}\right)$ $(\delta, \theta_1, \theta_2)$



RMS current of i_L → RMS current of T/F windings, RMS current of Switches

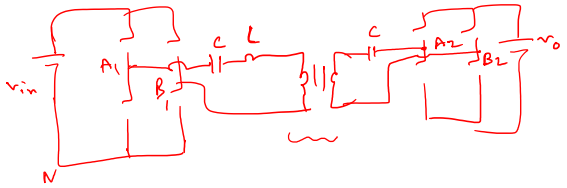
SPS, DPS, TPS, TPSVF, DPSVF

→ Resonant DAB converter

→ multilevel DAB converter

→ Modular Multilevel DAB converter

→ LLC resonant converter



AC-DC

isolated DC-DC → PSFB, DAB

Thank You